

$H \leq G$ subgroup of G .

G finite group

$$[G : H] = \# G/H$$

(number of distinct left cosets)

$$|H| = \text{cardinality of } H$$

$$|G| = [G : H] \cdot |H|$$

$$\left(\frac{|G|}{|H|} = [G : H] = |G/H| \right)$$

Conclude: if $H \leq G$, then

$$|H| \mid |G|$$

Today: Apply this to classify all groups of size 6

Lemma: if G is a finite group of even order, then there exists an element of order 2 in G .

Warning: Say $|G| = 24$

Possible subgroup sizes of G are:

1, 2, 3, 4, 6, 12, 24

But Lagrange's theorem does not tell you that you actually have a subgroup of each size

Proof of Lemma: For each $a \in G$, pair it with a^{-1}

$$\{a, a^{-1}\}$$

If a has order 2, then $a^{-1} = a$ since $a^2 = 1$

order of
group G
 $= \# \text{ elts of } G$

order of
 $x \in G$ is
smallest n
s.t. $x^n = 1$

OR

$\langle x \rangle = \text{subgp}$
generated by

$$x$$
$$= \{1, x, x^2, \dots, x^{-1}, x^{-2}, \dots\}$$

$$|\langle x \rangle| = \text{ord}(x)$$

Conversely if $a = a^{-1}$ ($a \neq 1$)

then $a^2 = 1$.

So: $(\{a, a^{-1}\} | = 2 \Leftrightarrow \text{ord}(a) = 2$
 $a \neq 1$

So now look at all elements of G .

Assume there is no element of order 2

So then the pairs $\{a, a^{-1}\}$ $a \neq 1$

always have size 2.

So every element $a \neq 1$ lives in exactly

one such pair, so how many elements does

G have?

$$|G| = \underbrace{1}_{a=1} + 2 \cdot \underbrace{k}_{\# \text{ of distinct pairs } \{a, a^{-1}\}}$$

$a=1$

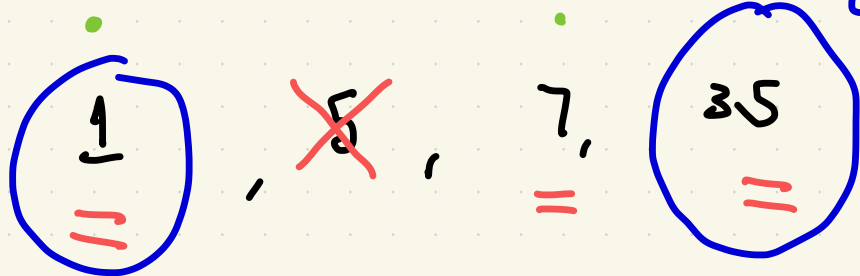
$\#$ of distinct pairs $\{a, a^{-1}\}$

$\Rightarrow |G|$ is odd, contradiction So
 one of the pairs $\{a, a^{-1}\}$ has
 size 1, i.e. $a^2 = 1$. $\square$

In homework: let $|G| = 35$, show
 that G has an element of order
 5 or 7

idea: Assume G has no elements of
 order 5, so all elements of G

have order



only one
 element of order 1 (e)

if $x \neq e$
 $x^{35} = 1$
 & no smaller
 power has is
 equal to 1
 (check: x^7 has
 order 5)

if every element has order 1 or 7, then

show that $|G| = 1 + 7k$.

So for: $|G| = 6$ then G has some element of order 2.

Extreme case: Suppose all elements of G satisfy

$$\boxed{x^2 = 1} \quad \text{for all } x \in G.$$

$\Rightarrow G$ is abelian!

Why? Given $a, b \in G$ want:

$$ab = ba.$$

Our assumption that $x^2 = 1$ all $x \in G$
i.e. for every $x \in G$, $x = x^{-1}$

$$\underline{b \cdot a} = b^{-1} \cdot a^{-1} = (\underline{a \cdot b})^{-1} = \underline{a \cdot b}$$

$$|G| = 6$$

possible orders of elements are

$$\underline{1}, \underline{2}, \textcircled{3}, \boxed{6} \leftarrow$$

$\Rightarrow$

Prop: If $|G| = 6$, then G has
to have an element x of order 3

Proof: Assume all elements of G
have order 2 (except for id), i.e.
every $x \in G$ satisfies $x^2 = 1$.

if G has element of order 6, it
automatically has element of order 3:

$$x^6 = 1 \Rightarrow (x^2)^3 = 1$$

& no smaller power = 1, $a = x^2$ $a^3 = 1$

$$a^2 = (x^2)^2 = x^4 \neq 1$$

So if all $x \in G$ satisfy $x^2 = 1$ then
 G is abelian.

Pick $a, b \in G$ order 2 $a, b \neq 1$
 $a \neq b$

$$K = \{1, a, b, ab\}$$

Claim: K is a subgroup!

Why? $a \cdot (a \cdot b) = a^2 b = b$

$$a^{-1} = a, \quad b^{-1} = b, \quad (ab)^{-1} = ab$$

$$ba \stackrel{?}{\in} K \quad \underline{ba} = \underline{ab}$$

so $|K| = 4$ but $|G| = 6$, contradiction!

So no such K exists, i.e.

every element in G satisfying $x^2 = 1$ is false!

Summarize:

$$|G| = 6$$

- identity 1
- has element y of order 2
- has element x of order 3

$$H = \{1, x, x^2\} \leq G.$$

Could $y \in H$? No: y has order 2.

So $y \notin H$.

Remember: coset $yH = H \iff y \in H$.

$$\therefore \text{ since } y \notin H, \quad yH \cap H = \emptyset$$

list $H = \{1, x, x^2\}$

$$yH = \{y, yx, yx^2\}$$

Now $|G| = 6$, so $G = H \cup yH$

i.e. $G = \{1, \underline{x}, \underline{x^2}, \underline{y}, \underline{yx}, \underline{yx^2}\}$

But what is the multiplication law on this set?

$$x^3 = 1, \quad y^2 = 1.$$

Claim: all we need to do is determine

What is xy ?

$$\cancel{xy} = \cancel{x}$$
$$y = 1.$$

Could $xy \in \{1, \underline{x}, \underline{x^2}\}$

$$\cancel{xy} = \cancel{x^2}$$

OR $\in \{y, yx, yx^2\}$

$$x \cancel{y} = \cancel{y} \quad \underline{\text{no!}} \quad \Rightarrow x = 1.$$

Only possibilities: $xy = \begin{cases} yx & (A) \\ yx^2 & (B) \end{cases}$

Fact: once we pick (A) or (B) the entire group multiplication law is determined by 3 rules: $\underline{x^3 = 1}$, $\underline{y^2 = 1}$, & $\begin{matrix} (A) - \\ \underline{\text{OR}} \\ (B) \end{matrix}$

e.s. Suppose

$$\underline{xy = yx}$$

$$G = \{1, x, x^2, y, yx, yx^2\}$$

Compute:

$$\underline{x^2} \cdot \underline{y} x^2 = ? \quad y \cdot x$$

$$x \cdot x \cdot y \cdot x \cdot x$$

$$y \cdot \underbrace{x \cdot x \cdot x \cdot x} = y \cdot x$$

(see notes for general proof)

So now it seems like we have at most 2 possibilities for group of order 6.

$$(A) \quad x^3 = 1, \quad y^2 = 1, \quad xy = yx$$

Abelian: this is just C_6

or cyclic group of order 6.

$$C_6 = \langle z \rangle \quad z^2 = x, \quad z^3 = y$$

$$(B) \quad x^3 = 1, \quad y^2 = 1, \quad \boxed{xy = yx^2}$$

$\Rightarrow$ the group S_3 !

$$\Rightarrow yx = x^2y$$

